

Where Limit Work Goes Wrong

Directions:

- Two errors account for most wrong limits: substituting when substitution is not allowed, and deciding an infinite limit from the numerator's sign alone.
- Before you compute anything, run the two checks: is the denominator zero at the target? and if so, what is the sign of the denominator on the side you are approaching from?
- In the find-and-fix problems, the printed work is a real student's. Give the correct answer *and* say what the mistake was — both count.

1. (a) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 + 2x}{x + 1}$ _____

(b) Explain, in one sentence, how you could tell before doing any arithmetic that direct substitution is safe for this limit.

2. **Find and fix.** A student wrote:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \frac{4 - 4}{2 - 2} = \frac{0}{0} = 0.$$

(a) Give the correct value: _____

(b) Name the error and state the step the student skipped.

3. Let $f(x) = \frac{1}{x-3}$. Track the sign of $x-3$ on each side of 3 and evaluate:

(a) $\lim_{x \rightarrow 3^-} f(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 3^+} f(x) = \underline{\hspace{2cm}}$

4. Evaluate $\lim_{x \rightarrow -1} \frac{x^2 + 3x + 2}{x^2 - 1}$ — direct substitution fails here, so rewrite first. Give the exact value.

Answer: $\underline{\hspace{2cm}}$

5. Find and fix. A student wrote:

“ $\lim_{x \rightarrow 2^-} \frac{x+4}{x-2}$: the top is heading to 6, which is positive, and the bottom is heading to 0, so the quotient blows up. The answer is $+\infty$.”

(a) Give the correct value: $\underline{\hspace{2cm}}$

(b) Name the error in the reasoning.

6. Let $g(x) = \frac{2-x}{(x-5)^2}$. The denominator is squared, so think about its sign carefully. Evaluate:

(a) $\lim_{x \rightarrow 5^-} g(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 5^+} g(x) = \underline{\hspace{2cm}}$

7. **Find and fix.** A student wrote:

“ $\lim_{x \rightarrow -2^-} \frac{x-6}{x+2}$: near $x = -2$ the top is about -8 , so the whole fraction is negative and grows without bound. The answer is $-\infty$.”

(a) Give the correct value: $\underline{\hspace{2cm}}$

(b) Name the error in the reasoning.

8. Challenge — find and fix. A student wrote:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{|x - 1|} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2.$$

(a) Compute the two one-sided limits:

$$\lim_{x \rightarrow 1^-} = \underline{\hspace{2cm}} \qquad \lim_{x \rightarrow 1^+} = \underline{\hspace{2cm}}$$

(b) There are two separate faults in the work above. Name both.